

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	03 October 2026
Team ID	SWTID-2026-8726
Project Name	Auto Ticket Classification using FLOW DESIGNER
Maximum Marks	4 Marks

Technical Architecture:

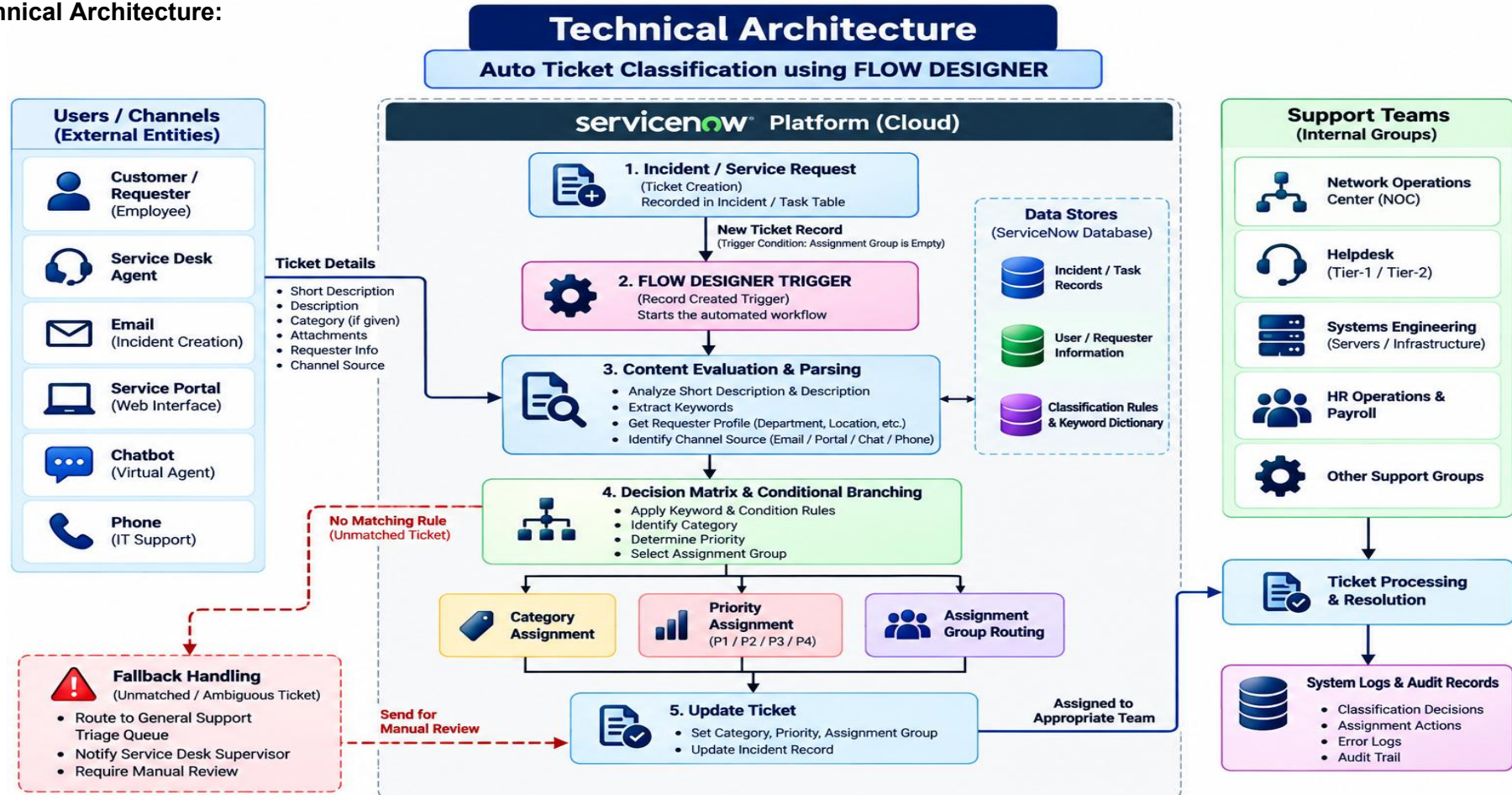


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	User Interface	Interface through which users submit incidents or service requests	ServiceNow Service Desk Portal / Web Interface
2. 2	Ticket Management	Creates and stores incoming incident/service request records	ServiceNow Incident / Task Table
3. 3	Workflow Automation	Automatically starts and controls the ticket-classification workflow	ServiceNow Flow Designer
4. 4	Content Evaluation	Evaluates ticket description, keywords, requester information and channel source	Flow Designer Data Processing / Flow Logic
5. 5	Classification Engine	Identifies ticket category using predefined conditions and keywords	Flow Designer If-Else / Switch Logic
6. 6	Priority Assignment	Determines appropriate P1–P4 priority according to configured conditions	Flow Designer Conditional Rules
7. 7	Assignment Routing	Automatically routes tickets to the appropriate support group	Flow Designer Update Record / Assignment Group Logic
8. 8	User Information	Provides requester department, location and other profile information	ServiceNow User / Requester Records
9. 9	Rule / Keyword Data	Stores classification keywords, conditions and routing mappings	ServiceNow Configuration / Flow Conditions
10. 10	Exception Handling	Handles unmatched, ambiguous and conflicting tickets	General Support Triage Queue / Flow Fallback Logic
11. 11	Logging & Audit	Records classification decisions, transformations and assignment actions	ServiceNow System Logs / Audit Records

The current project uses rule/keyword-driven Flow Designer automation. Machine Learning, sentiment analysis, multilingual support and dynamic load balancing are listed as future enhancements rather than current components.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No separate open-source application framework is specified in the current implementation.	ServiceNow Flow Designer
2.	Security	Workflow execution and ticket data access are controlled through the ServiceNow platform and configured application permissions.	ServiceNow Access Controls / Roles
3.	Scalable Architecture	Workflow automation reduces dependence on manual triage and can process increasing ticket volumes through automated classification and routing.	Flow Designer / ServiceNow Platform
4.	Availability	Automated workflow execution and fallback mechanisms help ensure that incoming tickets are not dropped when classification rules do not match.	Flow Designer + General Support Triage Queue
5.	Performance	Automatic classification and routing significantly reduce manual triage time. The project reports an average triage time reduction from 22.5 minutes to 1.2 seconds in its test results.	Flow Designer Automation

Final Technology Stack

Frontend / User Interface:
ServiceNow Service Desk Portal

Application / Workflow:
ServiceNow Flow Designer

Ticket Data:
Incident / Task Tables

Classification:
Keyword + Conditional Rule Engine

Decision Logic:
If-Else / Switch Branching

Routing:

Assignment Group Logic

User Data:

ServiceNow Requester/User Records

Exception Handling:

General Support Triage Queue

Logging:

System Logs & Audit Records

Infrastructure:

ServiceNow Enterprise Cloud Platform

Machine Learning:

Not in current implementation — Future Enhancement

This keeps the technology stack consistent with your actual project report instead of adding technologies like Python, React, MySQL or external APIs that your project documentation doesn't specify.